

Green skills education

What courses cost, how quality shows up in reviews and credentials, which skills are rising, and why a 4.7-star course still may not produce an installer.

This PDF is the full synthesis. You do not need GitHub, Python, or a spreadsheet. Snapshot dated 16 September 2026. Costs are typical unsubsidised list prices in US dollars unless a subsidy is named. FX: GBP 1.30, SGD 0.78, EUR 1.10, AUD 0.67.

1. Hiring is ahead of training Green hiring grew 7.7% vs 4.3% for green skills (LinkedIn 2025). About 20% of OECD workers are in green-driven jobs; only 2–14% of adult courses have green content.
2. Stars are not jobs England Heat Training Grant: 94% liked the course, 33% installed a heat pump, 12% newly got MCS. The bottleneck is a first install and a business accreditation, not another video.
3. Pay for the credential that employers use Solar jobs: NABCEP-class bootcamp (~\$895) beats a \$20 overview. Carbon jobs: GHG Protocol \$30–\$600 or GHGMI \$435 beat Harvard \$7,160 on value. Singapore: SkillsFuture can cut NTU list prices by 50–90%.

Contents: headlines · cost vs quality · cost bands · future skills · labour-market spend · heat-pump conversion · NTU Singapore · method and limits · full course list.

1. Headlines

Signal	Figure	Source
Green hiring vs green skills growth	7.7% vs 4.3% (still ~2x)	LinkedIn Green Skills 2025
Hiring premium for green talent	+46.6% vs overall workforce	LinkedIn 2025
Green hires in non-green job titles	53%	LinkedIn 2025
OECD workers in green-driven occupations	20%	OECD Employment Outlook 2024
Adult courses with green content	2.1–14.1% (AU, DE, SG, US)	OECD 2024
Employer skills changing by 2030	39%	WEF Future of Jobs 2025
Climate mitigation transforms the firm	47% of employers	WEF 2025
US solar PV installer median wage	\$53,140; jobs +37% 2025–35	BLS OOH
US wind turbine technician median wage	\$64,120; jobs +30% 2025–35	BLS OOH
US HVAC mechanic median wage	\$61,010; jobs +11% 2025–35	BLS OOH
UK employer spend per trainee	GBP 2,710 (series low)	DfE ESS 2024
Singapore SRA professionals	2,000 (2023) to 4,000 (2030)	Green Skills Committee 2025

Training supply is thin relative to job demand. Cheap MOOCs cover literacy. License and standard-setter courses are where employer-recognized quality concentrates.

Green hiring vs green skills

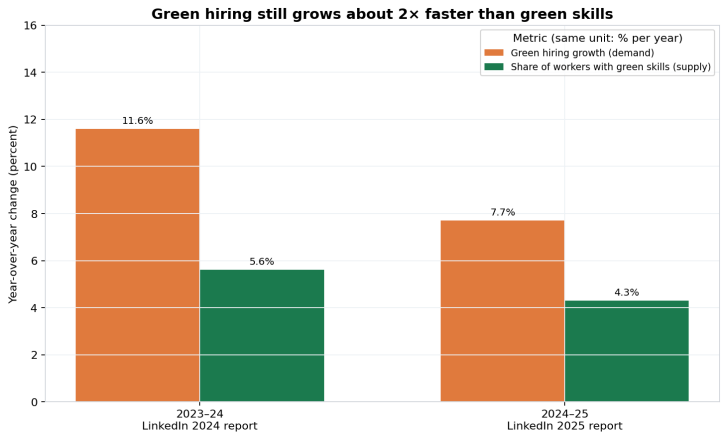
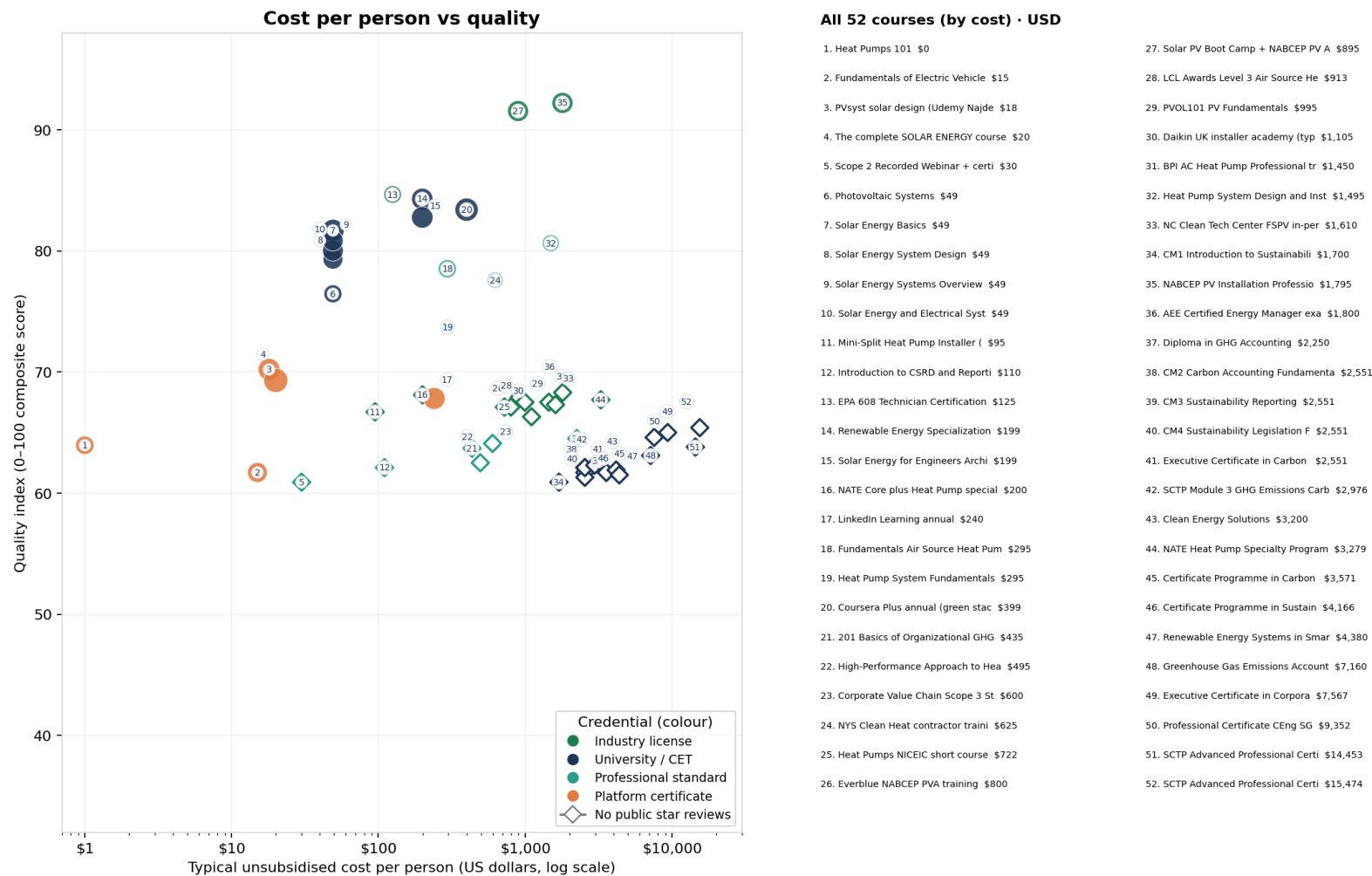


Figure 1. LinkedIn Economic Graph. Unit: percent year-over-year.

2. Cost per person vs quality

Quality index (0–100) = 35% star rating + 20% review-count confidence + 25% credential type + 20% employer recognition. A 5.0 with 8 reviews cannot beat a 4.7 with 1,000 reviews plus a license. Each number on the chart is one course; names and dollar amounts are in the key on the right. Diamonds have no public star ratings — quality is inferred from the credential.



Each number is one course (key at right). Filled circles = public star reviews (size = review count). Diamonds = no public stars. Quality = 35% stars + 20% review confidence + 25% credential + 20% employer signal.

Figure 2. Horizontal axis: unsubsidised cost per person, US dollars, log scale. Vertical axis: quality index. Colour: credential type.

Highest quality in this catalogue

Course	Provider	Cost USD	Stars	Reviews	Quality
NABCEP PV Installation Professional Certif	HeatSpring	\$1,795	4.72	1,080	92.2
Solar PV Boot Camp + NABCEP PV Associate E	HeatSpring / Sean White	\$895	4.72	1,046	91.5
EPA 608 Technician Certification	HeatSpring / Brynn Cooks	\$125	4.50	128	84.7
Renewable Energy Specialization	Coursera / University of	\$199	4.80	1,367	84.3
Coursera Plus annual (green stack access)	Coursera	\$399	4.70	5,000	83.4
Solar Energy for Engineers Architects and	Coursera / University at	\$199	4.70	1,487	82.8

Best value (quality divided by log cost)

Course	Cost USD	Quality	Value score
Heat Pumps 101	\$0	64.0	64.0
PVsyst solar design (Udemy Najdeah)	\$18	70.2	48.5
The complete SOLAR ENERGY course Beginner to advan	\$20	69.3	46.9
Solar Energy Basics	\$49	81.7	46.1
Solar Energy and Electrical System Design	\$49	80.8	45.6
Solar Energy Systems Overview	\$49	80.0	45.2

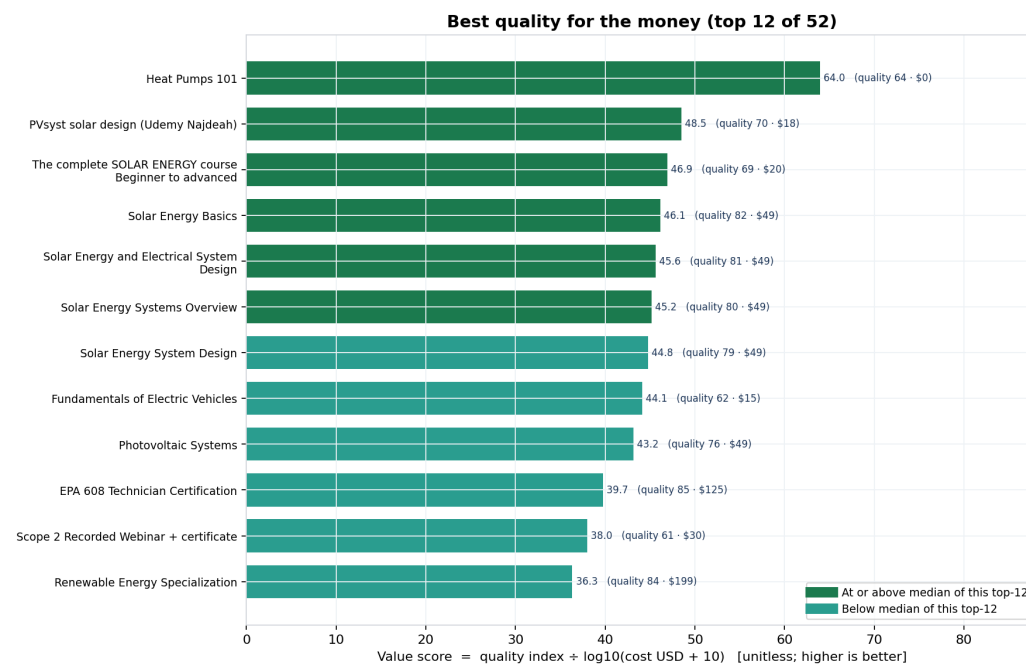


Figure 3. Top 12 value scores. Free literacy courses rank high because cost is zero.

3. Full cost ladder by skill cluster

Every course is here. Dollar labels are unsubsidised typical prices. Colour is credential type. UK and Singapore subsidies can cut the net price; those nets are discussed later, not on this chart.

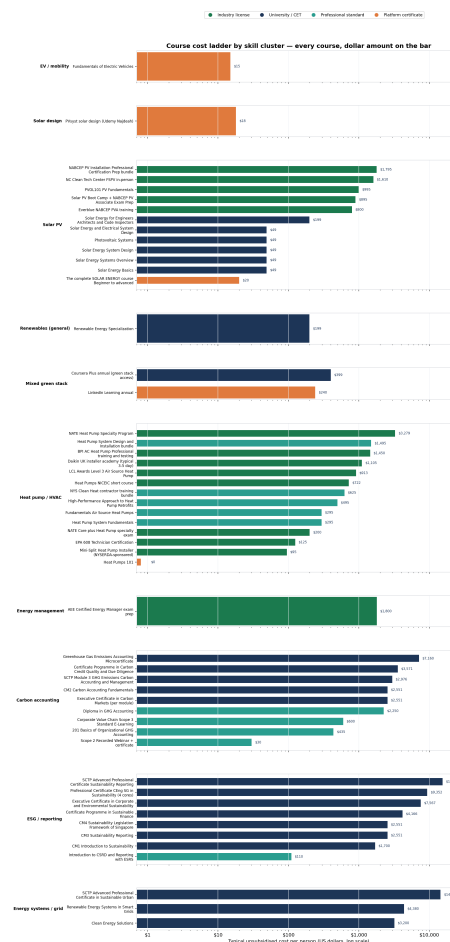


Figure 4. Log scale so a \$20 course and a \$15,000 certificate can share one axis.

4. What the price bands buy

Band	Typical USD / person	What you get	Review / quality pattern
Free / audit	0	Literacy, weak or no credential	High stars, low employer weight
Marketplace sale	13–25	Tool tutorials (PVsyst, solar, EV intro)	Many reviews; wide quality spread
University MOOC certificate	49–80	Single Coursera / edX certificate	4.6–4.8 stars, 100–2,600 reviews
Specialization / Plus year	199–399	Multi-course university stack	Strong ratings; mid-tier credential
Official standard e-learning	30–600	GHG Protocol Scope 2 (\$30) to Scope 3 (\$600)	Few public stars; high employer signal
Industry license bootcamp	800–1,800	NABCEP PVA/PVIP, AEE CEM, UK L3 ASHP	4.7 stars + pass rates where published
NTU Singapore classroom (list)	1,700–15,500	PACE / NBS CET, IES and ACRA pathways	No course-level stars; SkillsFuture 50–90% off
University microcertificate	7,000+	Harvard Extension GHG	Highest brand, weakest value in this set

Implication. For a solar installer job, the jump that matters is MOOC to NABCEP (~\$895), not Udemy to another Udemy. For carbon accounting, GHG Protocol \$30–\$600 or GHGMI \$435 beat Harvard \$7,160 on value unless the Harvard name is the product. A public voucher of about GBP 2,710 or EUR 3,000 covers one license bootcamp or a professional diploma — not a Harvard microcertificate, and not a scatter of \$15 courses.

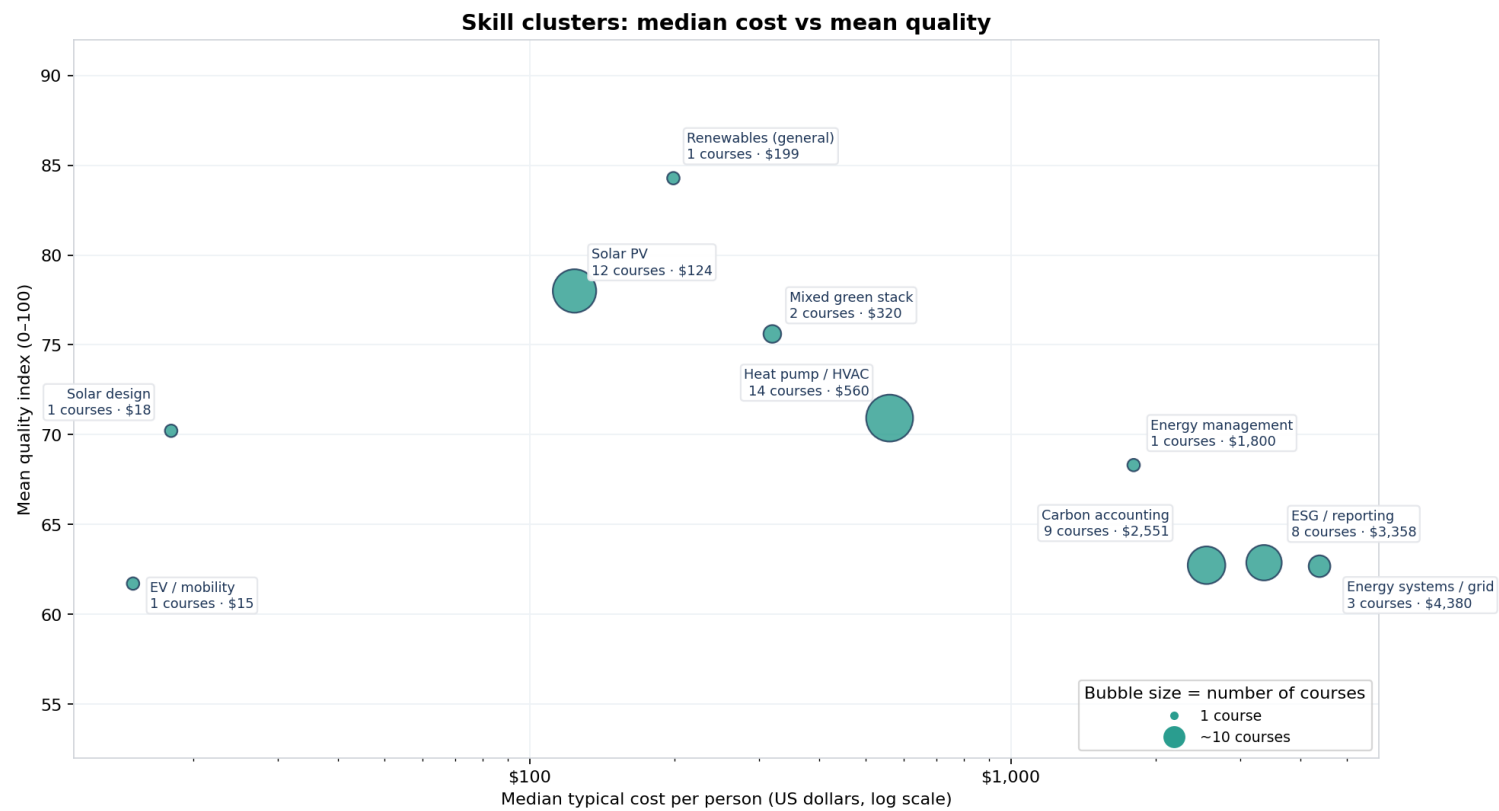


Figure 5. Skill clusters. Bubble size = number of courses in this catalogue.

5. Future courses worth stocking

Each panel below uses a different unit on purpose. Do not compare the 37% BLS job-growth bar with the 47% WEF employer bar as if they were the same thing.

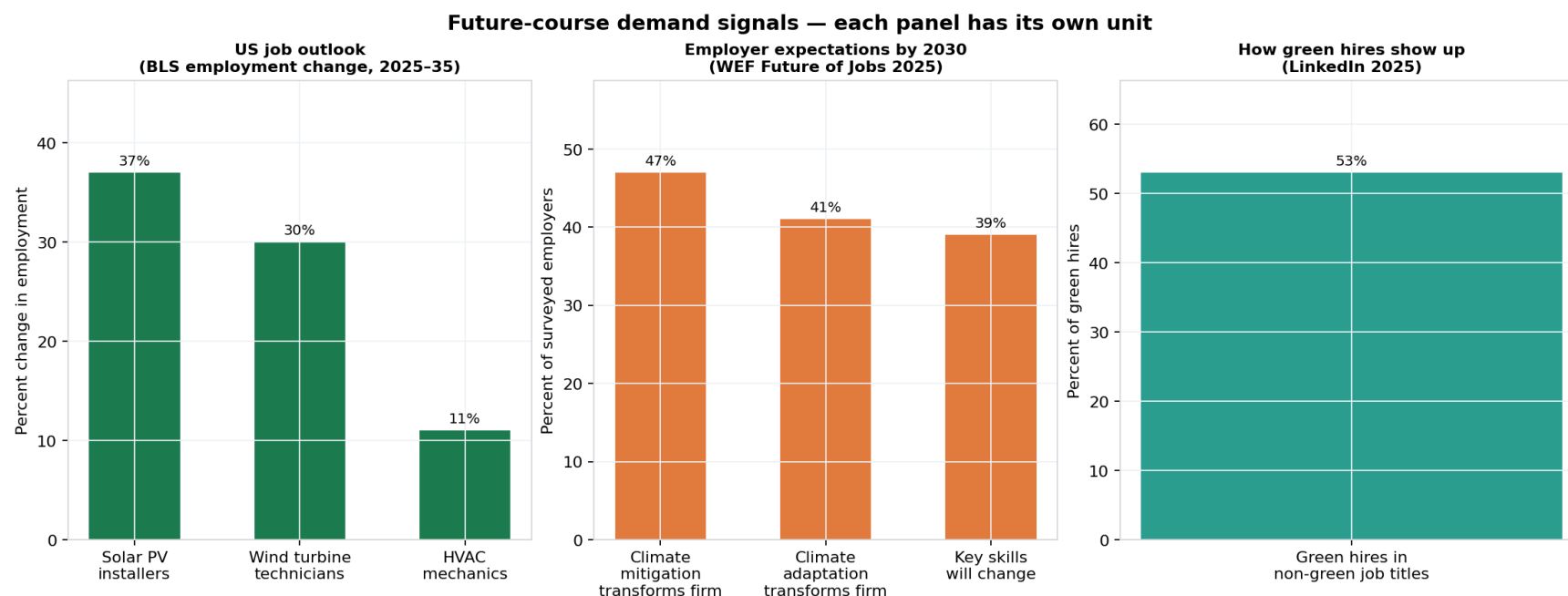


Figure 6. BLS = US employment change. WEF = share of employers. LinkedIn = share of green hires.

Priority inventory (demand times training gap)

- Heat-pump / HVAC electrification — UK ~100,000 engineers needed; US ~110,000 HVAC vacancies, heading toward ~225,000.
- Energy management — LinkedIn 2025 fastest-growing green skill (AI and data-centre load).
- Environmental stewardship — first time in the WEF top-10 growing skills.
- Solar PV install and design — BLS +37%; solar-design skills spiking on profiles.
- Wind turbine service — BLS +30%, median \$64,120.
- GHG / CSRD / ESRS / ISSB reporting — regulation-pulled; cheap intros and \$15k professional stacks coexist.
- EV / charging — WEF top-15 growing role.
- Grid flexibility and storage; sustainable procurement; green skills inside non-green jobs (53% of green hires).

6. Public spend vs wages

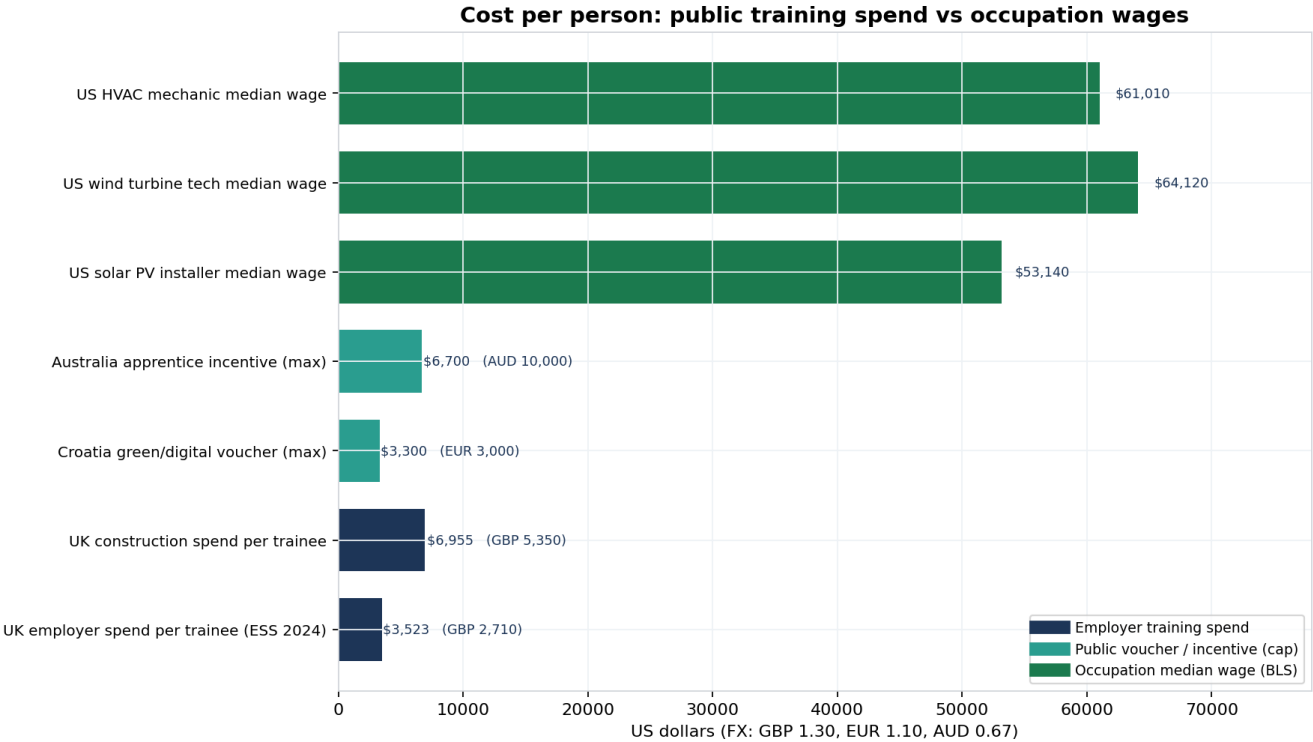


Figure 7. All bars in US dollars so they can be compared. Original currency in brackets.

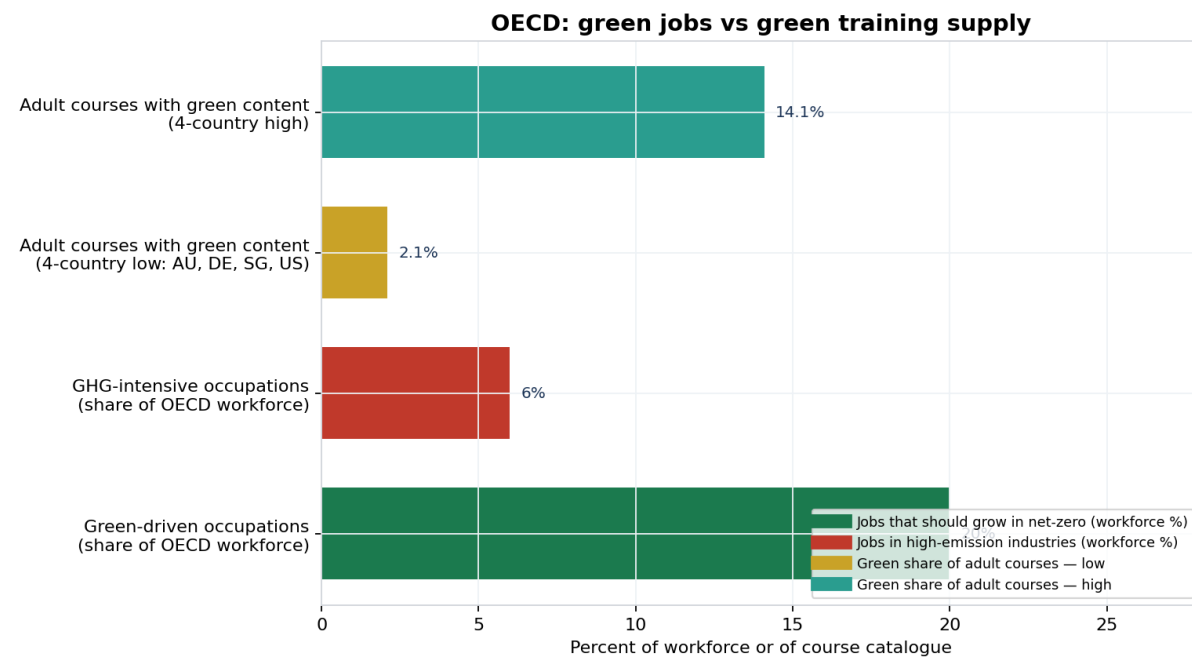


Figure 8. 20% of OECD workers are in green-driven occupations; 2–14% of catalogue courses are green.

Training cost against the job it points at

Each row is an occupation. Points on a row are courses aimed at that job, spread sideways so they do not sit on top of each other. Numbers match the key on the right. Bubble size is quality; colour is projected job growth.

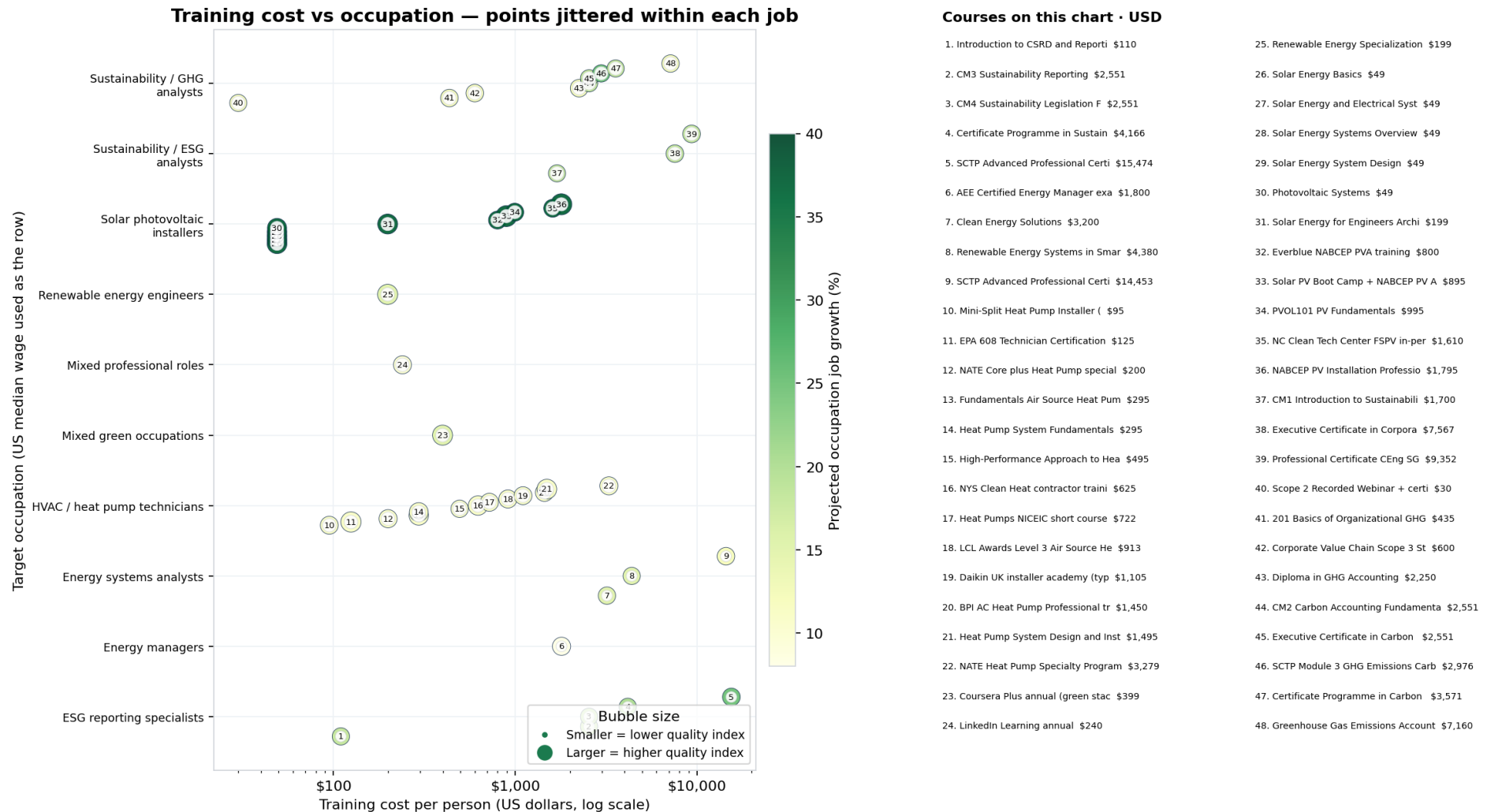


Figure 9. Wages are US BLS occupation medians, not course placement rates.

7. Heat pumps: reviews are not jobs

This is the highest-value finding in the pack. England's Heat Training Grant (up to GBP 500, 9,100 vouchers) is the rare dataset that measures both course satisfaction and whether people then install.

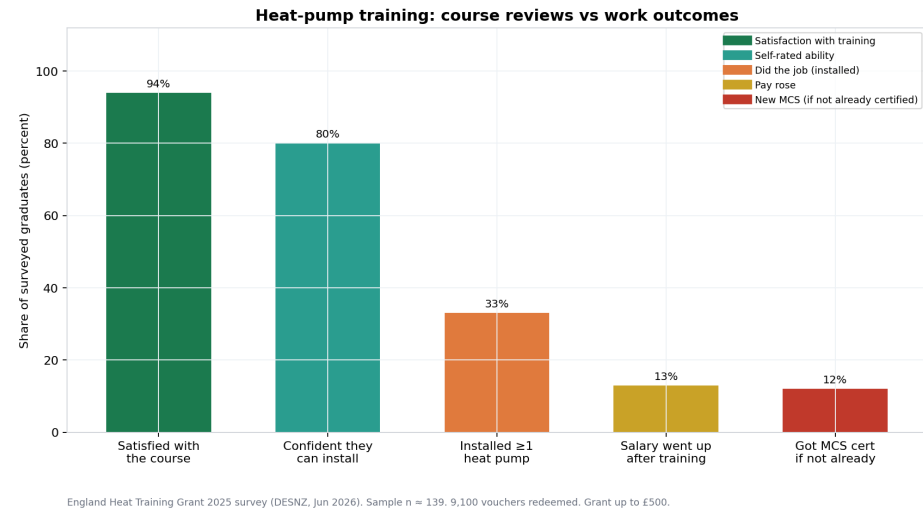


Figure 10. DESNZ HTG 2025 survey, n about 139.

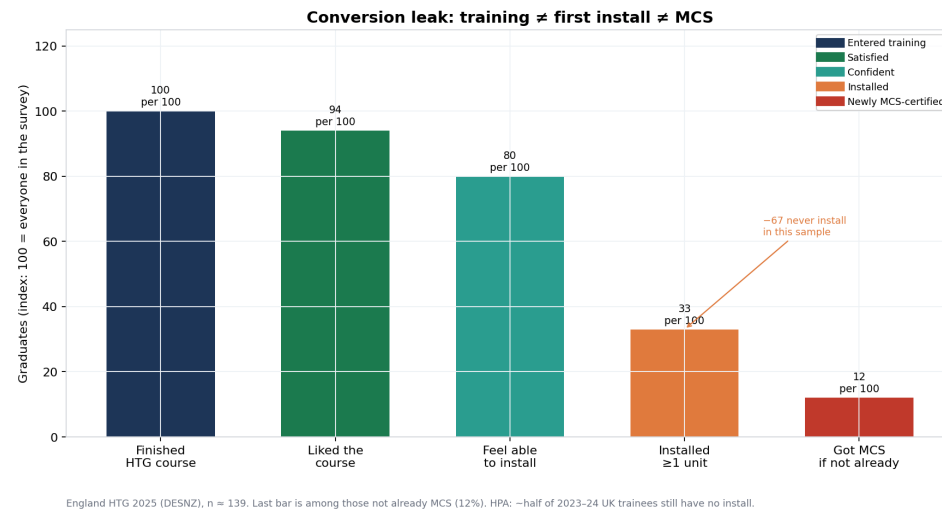


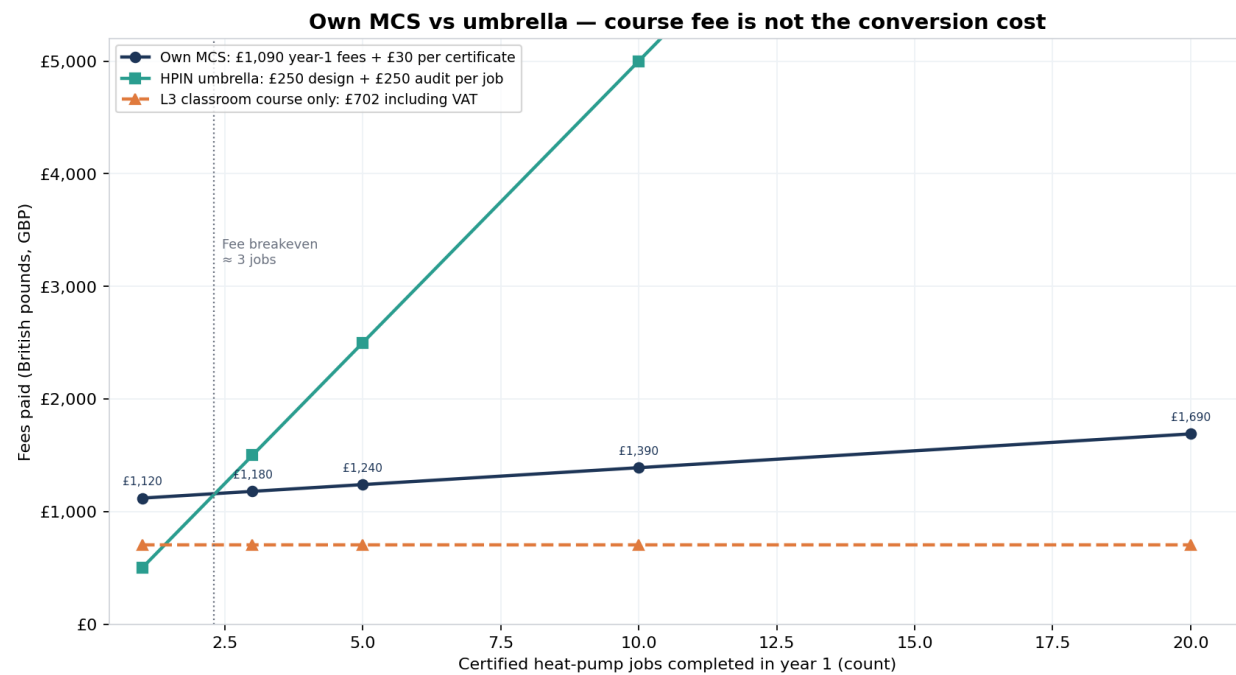
Figure 11. Per 100 graduates. Last bar is among those not already MCS-certified.

What actually converts a trainee into an installer

MCS is a business accreditation, not a personal exam. The Boiler Upgrade Scheme's GBP 7,500 grant only pays if the job is MCS-certified. The Heat Training Grant's GBP 500 pays for the course and stops there. HPA counted 7,800 UK course completions in 2023 and 7,000+ in 2024; about half still have no installation.

Path	GBP	USD at 1.30	What it does
L3 ASHP classroom (inc VAT)	702	913	The course HTG can discount
After Heat Training Grant	202	263	Net course in England
Own MCS year-1 fees (ex training)	1,090	1,417	Certification body + code + MCS
Course + own MCS year 1	~1,792	~2,330	What conversion actually costs
HPIN umbrella first job	500	650	They hold MCS; you install
Nesta Start at Home first install	0 kit	0	Funded kit in the engineer's own house

- Supervised first install — Nesta Start at Home: 2,000 registered, 250 live, 83 completed. Daikin adds GBP 750 off the first customer job.
- MCS umbrella for early jobs — HPIN (EDF): join GBP 0; GBP 250 design + GBP 250 audit per job. Same pattern at Alto, Baxi, manufacturer networks.
- Own MCS only if volume is coming — fee breakeven vs umbrella is about 3 certified jobs. After that own MCS is cheaper on cash. The real cost is QMS time, insurance, a first house, 2–4 months.



Own MCS £1,090 (Dwellow/MCS) excludes training time and insurance. Umbrella is charged every job. HTG £500 does not pay MCS fees. USD ≈ GBP × 1.30.

Figure 12. Vertical axis: British pounds. Horizontal axis: certified jobs in year 1.

8. NTU Singapore

Thirteen NTU PACE / Nanyang Business School offerings are in the catalogue. These are classroom continuing education with SkillsFuture funding, not MOOCs. There are no public course-level star ratings. MySkillsFuture rates NTU the provider at 4.2 from 16,427 reviews — institution-level, not copied onto each course. List prices in the charts are unsubsidised (same rule as the UK). Nets below are what a funded Singaporean actually pays.

NTU offering	List (inc GST)	SC 21–39 / PR	SC 40+ (MCES)
CM2 Carbon Accounting Fundamentals	S\$3,270 (~\$2,551)	S\$1,770 (~\$1,381)	S\$1,170 (~\$913)
SCTP GHG module	S\$3,815 (~\$2,976)	S\$1,145 (~\$893)	S\$445 (~\$347)
SCTP Sustainability Reporting + AI (135h)	S\$19,838 (~\$15,474)	S\$5,951 (~\$4,642)	S\$2,311 (~\$1,803)
Sustainable Finance certificate	S\$5,341 (~\$4,166)	S\$2,891 (~\$2,255)	S\$1,911 (~\$1,491)
Carbon Markets exec (per module)	S\$3,270 (~\$2,551)	S\$1,770	S\$1,170
Renewable Energy Systems in Smart Grids	~S\$5,616 (~\$4,380)	~S\$1,685 (70%)	SME ~S\$654
CEng (SG) Sustainability 4 cores	S\$11,990 (~\$9,352)	—	electives still TBC

Singapore demand signal	Figure	Source
Sustainability reporting professionals	2,000 (2023) → 4,000 (2030)	Green Skills Committee 2025
Solar / storage / smart grid / power-import workforce	+70% to ~720 by 2026	Green Skills Committee 2025
Green-jobs demand growth 2024	+27% (fastest in Asia)	LinkedIn
Carbon Markets Academy at NTU	300 professionals by 2027	EDB / NTU
Green workforce budget	S\$235 million	SBR 2026

GHGMI 201 is \$435 with an exam. NTU CM2 is \$2,551 list / about \$913 after 70% funding. You are paying for classroom time, the IES Chartered Engineer (SG) pathway, and Singapore statute — not the same product as a \$435 e-learn. After MCES, NTU carbon accounting sits near GHG Protocol Scope 3 (\$600) and well below Harvard (\$7,160). The SCTP reporting certificate at \$15k list is the ISSB / ACRA compliance stack, not a solar-installer bootcamp.

9. How quality was scored, and the limits

$\text{quality_index} = 0.35 \times (\text{stars} / 5 \times 100) + 0.20 \times (\log_{10}(\text{reviews}+1) / 4 \times 100) + 0.25 \times \text{credential} + 0.20 \times \text{employer signal}$. Missing stars are scored as a neutral 70, and review confidence is then zero, so unreviewed professional courses are not fake-4.8'd and not crushed to zero.

What reviews can and cannot do

- They measure learner satisfaction (clarity, production, instructor).
- They do not measure job placement. HeatSpring's 88% NABCEP pass rate and the HTG 33% install rate are better outcome metrics.
- A marketplace 4.4 with 11,000 reviews is a popularity signal, not an employer signal.

Limits of this snapshot

- Udemy list prices are fictional for most buyers; sale price is used as typical.
- Coursera specializations assume about 4 months at \$49/month unless the row is Coursera Plus (\$399/year).
- GHG Protocol, GHGMI, AEE, SEI, NATE programmes, and NTU courses often publish no course-level stars.
- One EV row is low-confidence (estimated reviews).
- Wages are US occupation medians, not graduate placement. HVAC wage is the heat-pump proxy (no separate BLS heat-pump installer code).
- LinkedIn green skills are self-reported profile skills.
- HTG survey n is about 139 — directional, not a census.
- This is not legal, careers, or investment advice. Confirm live prices and funding rules before buying a course.

10. Full course list (all 52)

Sorted by cluster then cost. Quality is the composite 0–100 index. A dash in Stars / Reviews means no public rating was found; quality then leans on credential and employer signal.

Course	Provider	Cluster	Cost USD	Stars	Reviews	Quality
Scope 2 Recorded Webinar + certificate	GHG Protocol / WRI	carbon accounting	\$30	—	—	61
201 Basics of Organizational GHG Accounting	GHG Management Institute	carbon accounting	\$435	—	—	64
Corporate Value Chain Scope 3 Standard E-Learnin	GHG Protocol / WRI	carbon accounting	\$600	—	—	64
Diploma in GHG Accounting	GHG Management Institute	carbon accounting	\$2,250	—	—	64
CM2 Carbon Accounting Fundamentals	NTU PACE	carbon accounting	\$2,551	—	—	62
Executive Certificate in Carbon Markets (per mod	NTU NBS / Asian School of th	carbon accounting	\$2,551	—	—	62
SCTP Module 3 GHG Emissions Carbon Accounting an	NTU Nanyang Business School	carbon accounting	\$2,976	—	—	62
Certificate Programme in Carbon Credit Quality a	NTU Nanyang Business School	carbon accounting	\$3,571	—	—	62
Greenhouse Gas Emissions Accounting Microcertifi	Harvard Extension School	carbon accounting	\$7,160	—	—	63
AEE Certified Energy Manager exam prep	AEE / partners	energy management	\$1,800	—	—	68
Clean Energy Solutions	MIT Professional Education	energy systems	\$3,200	—	—	63
Renewable Energy Systems in Smart Grids	NTU PACE / EEE	energy systems	\$4,380	—	—	62
SCTP Advanced Professional Certificate in Sustai	NTU PACE	energy systems	\$14,453	—	—	64
Introduction to CSRD and Reporting with ESRS	GRI Academy	esg reporting	\$110	—	—	62
CM1 Introduction to Sustainability	NTU PACE	esg reporting	\$1,700	—	—	61
CM3 Sustainability Reporting	NTU PACE	esg reporting	\$2,551	—	—	62
CM4 Sustainability Legislation Framework of Sing	NTU PACE	esg reporting	\$2,551	—	—	61
Certificate Programme in Sustainable Finance	NTU Nanyang Business School	esg reporting	\$4,166	—	—	62
Executive Certificate in Corporate and Environme	NTU Nanyang Business School	esg reporting	\$7,567	—	—	65
Professional Certificate CEng SG in Sustainabili	NTU PACE / IES	esg reporting	\$9,352	—	—	65

Course	Provider	Cluster	Cost USD	Stars	Reviews	Quality
SCTP Advanced Professional Certificate Sustainab	NTU Nanyang Business School	esg reporting	\$15,474	—	—	65
Fundamentals of Electric Vehicles	Udemy	ev mobility	\$15	4.30	500	62
Heat Pumps 101	HeatSpring / John Siegenthal	heat pump hvac	\$0	4.77	214	64
Mini-Split Heat Pump Installer (NYSERDA-sponsore	ITEC Training	heat pump hvac	\$95	—	—	67
EPA 608 Technician Certification	HeatSpring / Brynn Cooksey	heat pump hvac	\$125	4.50	128	85
NATE Core plus Heat Pump specialty exam	NATE	heat pump hvac	\$200	—	—	68
Fundamentals Air Source Heat Pumps	HeatSpring	heat pump hvac	\$295	4.70	174	79
Heat Pump System Fundamentals	HeatSpring / John Siegenthal	heat pump hvac	\$295	4.70	22	74
High-Performance Approach to Heat Pump Retrofits	NCI / TECH Clean California	heat pump hvac	\$495	—	—	62
NYS Clean Heat contractor training bundle	HeatSpring	heat pump hvac	\$625	4.68	57	78
Heat Pumps NICEIC short course	Crawley College	heat pump hvac	\$722	—	—	67
LCL Awards Level 3 Air Source Heat Pump	Logic4training	heat pump hvac	\$913	—	—	68
Daikin UK installer academy (typical 3-5 day)	Daikin Training Centre	heat pump hvac	\$1,105	—	—	66
BPI AC Heat Pump Professional training and testi	EnergyScore	heat pump hvac	\$1,450	—	—	68
Heat Pump System Design and Installation bundle	HeatSpring	heat pump hvac	\$1,495	4.82	94	81
NATE Heat Pump Specialty Program	HVACRedu.net	heat pump hvac	\$3,279	—	—	68
LinkedIn Learning annual	LinkedIn Learning	mixed green	\$240	4.40	2,000	68
Coursera Plus annual (green stack access)	Coursera	mixed green	\$399	4.70	5,000	83
Renewable Energy Specialization	Coursera / University of Col	renewables	\$199	4.80	1,367	84
PVsyst solar design (Udemy Najdeah)	Udemy / Noor Aldeen Najdeah	solar design	\$18	4.60	2,000	70
The complete SOLAR ENERGY course Beginner to adv	Udemy / Leandro Caruso	solar pv	\$20	4.40	11,126	69
Solar Energy Basics	Coursera / SUNY	solar pv	\$49	4.80	2,600	82
Solar Energy and Electrical System Design	Coursera / University at Buf	solar pv	\$49	4.70	733	81

Course	Provider	Cluster	Cost USD	Stars	Reviews	Quality
Solar Energy Systems Overview	Coursera / University at Buf	solar pv	\$49	4.70	874	80
Solar Energy System Design	Coursera / SUNY	solar pv	\$49	4.70	480	79
Photovoltaic Systems	Coursera / DTU	solar pv	\$49	4.60	161	76
Solar Energy for Engineers Architects and Code I	Coursera / University at Buf	solar pv	\$199	4.70	1,487	83
Everblue NABCEP PVA training	Everblue	solar pv	\$800	—	—	67
Solar PV Boot Camp + NABCEP PV Associate Exam Pr	HeatSpring / Sean White	solar pv	\$895	4.72	1,046	92
PVOL101 PV Fundamentals	Solar Energy International (	solar pv	\$995	—	—	68
NC Clean Tech Center FSPV in-person	NC Clean Tech Center	solar pv	\$1,610	—	—	67
NABCEP PV Installation Professional Certificatio	HeatSpring	solar pv	\$1,795	4.72	1,080	92

Primary sources include OECD Training Supply for the Green and AI Transitions (2024), OECD Employment Outlook 2024, WEF Future of Jobs 2025, LinkedIn Green Skills Reports 2024–2025, BLS Occupational Outlook Handbook (May 2025 wages), UK DfE Employer Skills Survey 2024, DESNZ Heat Training Grant 2025 mid-scheme review, Nesta Start at Home, MCS / NICEIC fee tables, Singapore Green Skills Committee Report 2025, and live NTU PACE / NBS fee pages. Course ratings from Coursera, Udemy, HeatSpring, and MySkillsFuture as dated in the underlying tables.

If someone later needs the spreadsheets or to refresh the numbers, the same briefing lives with the data files. This PDF is meant to be forwarded on its own.